

Mission Team Communication Protocol

Design and test a concise communication protocol for a multi-station engineering task.

Intermediate	1 class period	Aerospace Mission
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Engineering Context

A rover test team must coordinate a driver, sensor observer, data recorder, and safety lead. Miscommunication could damage hardware or invalidate test data.

What You Will Practice

- Define clear team roles
- Create standardized callouts
- Use closed-loop communication
- Evaluate communication performance

What You Need

- Role cards
- Simple timed team task
- Communication log
- Engineering notebook

Student Instructions

- 1 Assign four team roles and define each role's authority.
- 2 Write standard callouts for ready, start, stop, fault, data confirmed, and reset.
- 3 Create a rule requiring the receiver to repeat critical commands.
- 4 Run a baseline task using normal conversation and record errors or delays.
- 5 Run the task again using the protocol.
- 6 Compare completion time, errors, and missed information.
- 7 Revise two parts of the protocol.

Engineering Requirements

- Every critical command must receive confirmation
- One person must have stop-work authority
- Protocol must distinguish normal completion from emergency stop
- Testing must include before/after evidence

Constraints & Safety

- Use a low-risk instructor-approved task
- No running or unsafe movement
- The safety lead may stop the task at any time

What You Submit

- Role chart
- Communication protocol
- Baseline and final performance table
- Revision notes
- Team reflection

Reflection Questions

- 1 Which callout prevented the most confusion?
- 2 Did the protocol slow the team down or make it faster overall?
- 3 What information should never be assumed?
- 4 How could the protocol work in a noisy environment?

Engineering Tip

Use short, unique phrases that cannot be confused with casual conversation.

Common Mistake

Adding too many callouts can make the system harder to use. Standardize only critical information.

Stretch Goal

Add a hand-signal or visual backup for two critical commands.

Career Connection

Flight-test and launch teams use disciplined communication because ambiguous instructions can create serious risk.